6. AI Integration Workflow

6.1 Why AI is Needed

The AI-Based Smart Quiz System incorporates artificial intelligence to address two fundamental challenges faced in traditional quiz management and examination environments.

First, manual question creation is a time-consuming and resource-intensive process. Teachers are required to individually compose each question, formulate four plausible answer options, identify the correct response, and write an explanatory note — all for every topic they wish to assess. In a course with a broad syllabus, this places a significant burden on the instructor and limits the volume and variety of questions that can be realistically produced. Automating this process through AI enables teachers to generate complete, well-structured MCQs in seconds by simply providing a topic keyword.

Second, detecting academic dishonesty during online quizzes is difficult to accomplish manually. When students take quizzes in an unproctored environment, behaviours such as rapidly switching browser tabs to consult external resources or completing questions at an unnaturally fast pace are difficult to identify without automated assistance. The system addresses this by integrating client-side behavioural monitoring to detect and flag suspicious quiz attempts, thereby preserving the integrity of the assessment process.

6.2 AI Service and Model Used

AI Question Generation: The system integrates with the Google Gemini API, using the model gemini-2.0-flash. This model is selected for its speed, efficiency, and ability to produce well-formed, contextually accurate multiple-choice questions from a brief topic prompt. Communication with the Gemini API is handled server-side within the Laravel application to ensure API key security and consistent response parsing.

Cheating Detection: The cheating detection mechanism does not rely on an external AI service. It is implemented using custom JavaScript logic embedded in the quiz page, combined with server-side validation in Laravel. The JavaScript visibilitychange event tracks tab-switching behaviour, while the server calculates average time per question from the submitted quiz timestamps. This rule-based logic effectively identifies behavioural anomalies without requiring a third-party AI provider.

6.3 Input to the AI System

For the AI Question Generation feature, the teacher provides a single input: a topic string. This is a short text phrase describing the subject area for which the MCQ should be generated (for example, “4G LTE Architecture” or “Bluetooth Protocol”). The teacher enters this topic into a dedicated input field on the question management page for the Mobile Computing Lab course. The Laravel controller then constructs a structured prompt around this topic and submits it to the Google Gemini API.

6.4 Output from the AI System

The Google Gemini API returns a response that the Laravel backend parses into a structured JSON object. The expected output format is as follows:

```
{
   "question": "What is the primary function of the MAC layer in a mobile network?",
   "options": [
      "A. To manage physical radio transmission",
      "B. To control medium access and scheduling",
      "C. To assign IP addresses to mobile devices",
      "D. To encrypt application-layer data"
   ],
   "correct_answer": "B",
   "explanation": "The MAC (Medium Access Control) layer is responsible for controlling how devices access the shared radio medium, including scheduling and resource allocation."
}
```

This JSON response is returned to the front-end via the Laravel controller, and the values are automatically populated into the corresponding form fields — question text, four option inputs, the correct answer selector, and the explanation field — allowing the teacher to review, adjust if necessary, and save the question.

6.5 How AI Improves the Project

The integration of AI into the Smart Quiz System delivers the following benefits:

- Efficiency: Teachers can generate a complete, ready-to-use MCQ within seconds by entering a single topic keyword, eliminating the need to manually compose questions, options, and explanations from scratch. This significantly reduces question preparation time.
- Quality and Consistency: The Gemini model produces grammatically correct, contextually relevant questions with clearly differentiated answer options and accurate explanations, maintaining a consistent standard of question quality across all AI-generated content.
- Scalability: The system can support a large and growing question bank without placing additional workload on teachers. As course content expands, AI generation allows rapid creation of questions covering new topics.
- Academic Integrity: The automated cheating detection mechanism flags attempts involving tab switching or abnormally fast responses, allowing the system to exclude suspicious submissions from the leaderboard and providing teachers with an audit trail of flagged attempts for review.
- Improved Student Experience: Students benefit from a richer and more varied question pool, receive immediate explanations after each quiz, and participate in a fair and integrity-protected assessment environment.

6.6 AI Question Generation Workflow

The following steps describe the end-to-end workflow for AI-assisted question generation in the system:

1. The teacher logs into the system with their credentials and navigates to the question management page for the Mobile Computing Lab course.
2. The teacher clicks the “Generate Question with AI” button, which reveals a topic input field on the page.
3. The teacher types a topic string (e.g., “OFDM in 4G Networks”) into the input field and submits the request.
4. The browser sends a POST request to the Laravel route /teacher/generate-ai-question, passing the topic string and the course identifier in the request body.
5. The Laravel controller receives the request, constructs a structured natural-language prompt incorporating the topic, and sends it to the Google Gemini API (model: gemini-2.0-flash) using an authenticated HTTP request with the server-stored API key.
6. The Gemini API processes the prompt and returns a JSON-formatted response containing the question text, an array of four answer options, the correct answer identifier, and an explanation.
7. The Laravel controller parses the API response and returns the structured JSON data to the browser.
8. JavaScript on the question form page receives the JSON response and automatically populates the form fields: question text, option A through D, correct answer, and explanation.
9. The teacher reviews the auto-filled question, makes any necessary edits, and clicks “Save” to persist the question to the database via the standard POST /teacher/courses/{course_id}/questions route.
10. The question is saved and becomes immediately available for inclusion in student quizzes for the Mobile Computing Lab course.

6.7 Workflow Diagram :